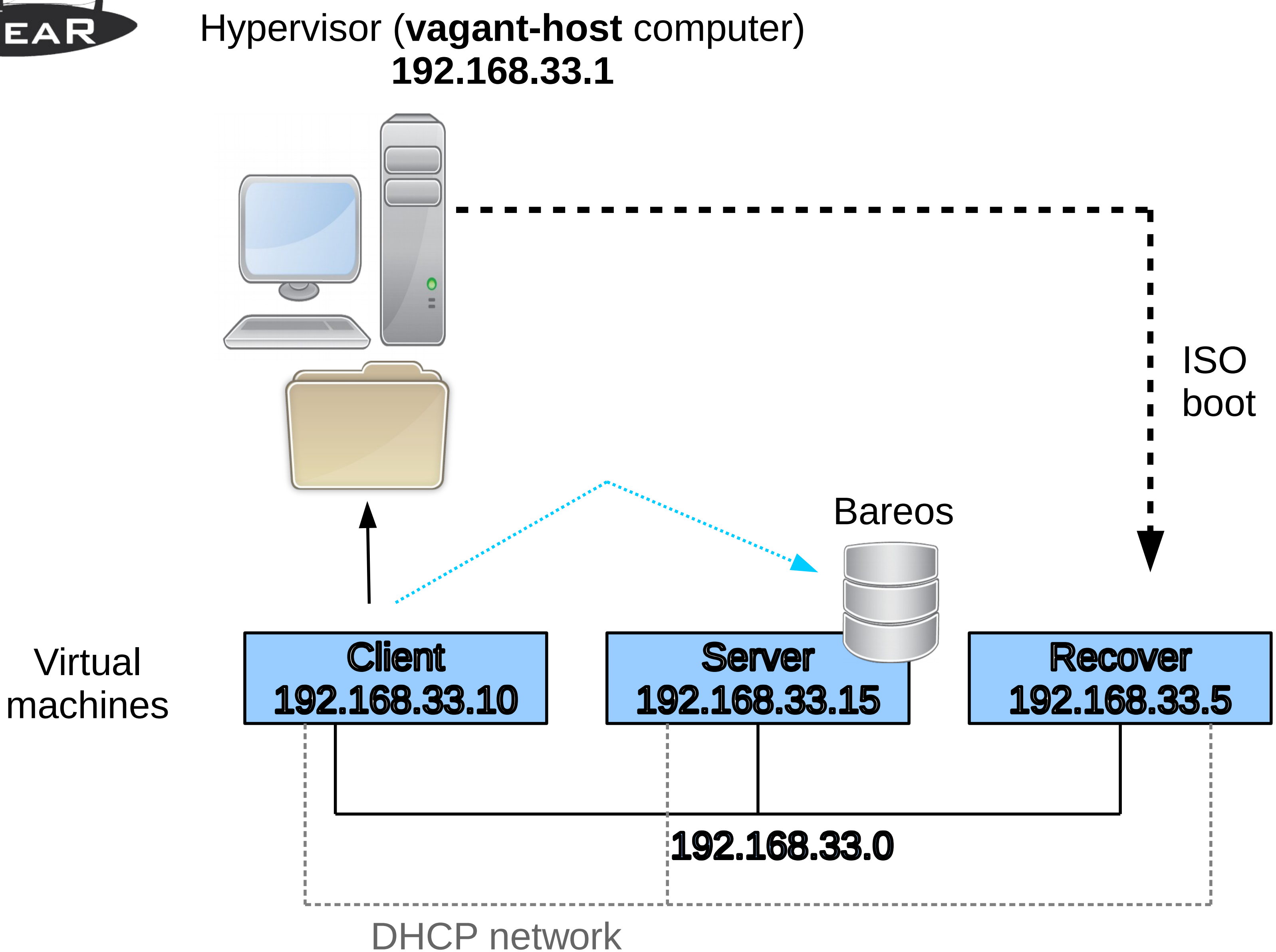
Relax and Recover

Open Source Backup Conference

Lab Exercises of the workshop

Gratien D'haese
IT3 Consultants

Lab Overview



Lab 1

Set up vagrant environment

Set up vagrant environment

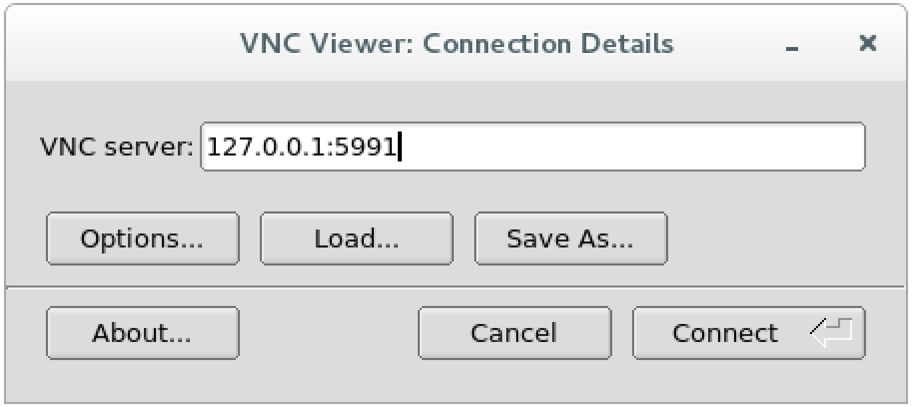
- Host system can be Linux, Mac, Windows
- A hypervisor like KVM, VirtualBox, VMware Player or VMware Fusion, Parallels Desktop
- Install “vagrant” from your distro, or from https://www.vagrantup.com/downloads.html
- KVM with libvirt needs the vagrant-libvirt plugin
  vagrant plugin install vagrant-libvirt
- Install “git” software to clone the Vagrantfile and scripts

Set up vagrant environment (2)

- git clone git@github.com:rear/rear-workshop-osbconf-2016.git
- Go into directory “centos7”
- Type “vagrant up --provision”
- Use “vagrant ssh client” to login on the client system
- Use “vagrant ssh server” to login on the server system
- Account vagrant/vagrant (and root/vagrant)

Set up vagrant environment (3)

- If you install “tigervnc” you can use vncviewer
- Use address 127.0.0.1 (localhost)
- Port 5991 for “client”
- Port 5992 for “server”
- Port 5993 for “recover”



Set up vagrant environment (4)

- Another way to login is via ssh:
  - ssh root@192.168.33.10 (client root pw is vagrant)
  - ssh root@192.168.33.15 (server root pw is vagrant)

Lab 2

Exchange root's public SSH key

Exchange root's public SSH key

- Login on client as root (via vagrant user)
- Create a SSH keypair with “ssh-keygen -t rsa”
- Secure copy to “server” and *append* it to /root/.ssh/authorized_keys, or use

  “ssh-copy-id root@server”
- If you want to use the “vagrant-host” as destination for the ISO image then repeat above action
- Test if it work – use “ssh server ls”

Lab 3

Install relax-and-recover

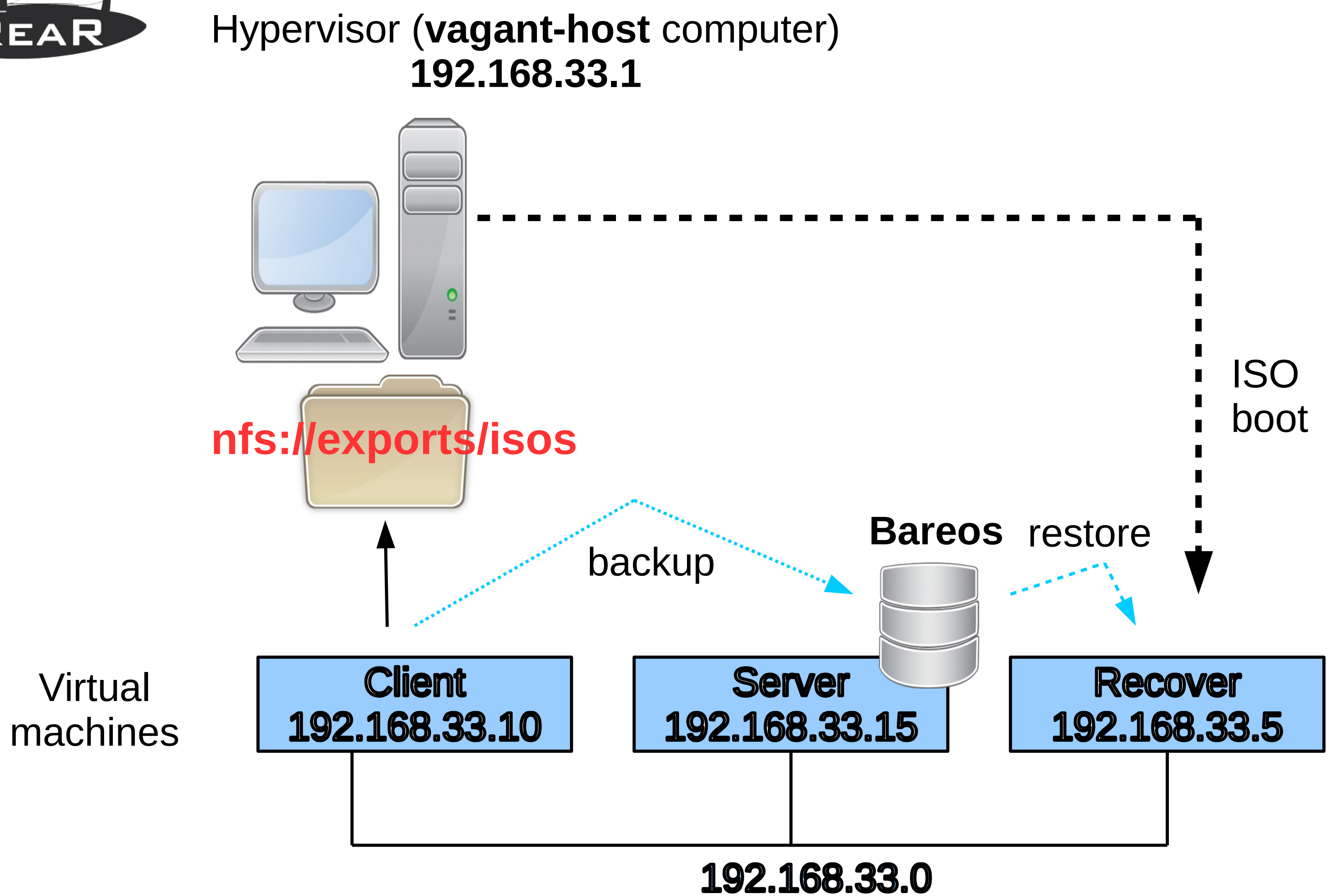
Install relax-and-recover (1)

- yum search rear
- yum -y install rear
- Install some extra tools:
  - yum -y install net-tools (for netstat, route,...)
  - yum -y install bind-utils (for nslookup, dig,...)
  - yum -y install rear-workshop (for the configs)

Lab 4

Make a full backup with Bareos

Lab 4: using Bareos



Make a full backup with Bareos

- Login on client
- Start the “bconsole”
  * run
  Automatically selected Catalog: MyCatalog

  The defined Job resources are:
  1: client-backup
  2: client-restore
  3: client-backup-mysql
  4: client-restore-mysql
  Select Job resource (1-4): 1

Configure relax-and-recover

- Configure /etc/rear/local.conf as follow:
```
OUTPUT=ISO
OUTPUT_URL=nfs://vagrant-host/exports/isos
ISO_DEFAULT=manual
BACKUP=BAREOS
USE_STATIC_NETWORKING=y
```
- Or, copy the prepared configuration file :
```
cp /etc/rear/workshop/local-with-bareos.conf /etc/rear/local.conf
```
- Run “rear -v dump”

Lab 5

Create rear rescue image for BACKUP=BAREOS topology

Create rear rescue image

- On <u>client</u>
- Run “rear -v mkrescue”
- Fix any error shown (BAREOS_RESTORE_JOB)
- Verify /var/log/rear/rear-client.log
- Check destination (OUTPUT_URL) host for ISO
- Is NFS Server properly configured?
  - Is /exports/isos exported?
  - Firewall/iptables rules?

Problems with NFS server?

- An excellent aid is the following site:
  http://www.server-world.info/en/note?os=CentOS_7&p=nfs&f=1
- $ cat /etc/exports
  /exports 192.168.0.0/16(rw,no_root_squash)

Create rear rescue image

- Typical output is:

```
# rear -v mkrescue
Relax-and-Recover 1.18-git201606091742
Using log file: /var/log/rear/rear-client.log
Creating disk layout
Creating root filesystem layout
Copying files and directories
Copying binaries and libraries
Copying kernel modules
Creating initramfs
Making ISO image
Wrote ISO image: /var/lib/rear/output/rear-client.iso
Copying resulting files to nfs location
```

Recover with Bareos

- Halt the client on vagant-host: vagrant halt client
- Verify on the vagrant-host that the /exports/isos/client is readable for everyone
- Start the recover: vagrant up recover
- Connect with vncviewer to 127.0.0.1:5993
- Login with root
- Recover your 'client' system onto vm 'recover'
  - Check network with 'ip -a'
  - Recover with 'rear -v recover'

Recover with Bareos (2)

QEMU (centos7_recover) - TigerVNC

```
Original disk /dev/vda does not exist in the target system. Please choose an app
ropriate replacement.
1) /dev/vda
2) Do not map disk.
#? 1
2016-06-28 13:21:18 Disk /dev/vda chosen as replacement for /dev/vda.
Disk /dev/vda chosen as replacement for /dev/vda.
This is the disk mapping table:
    /dev/vda /dev/vda
Please confirm that '/var/lib/rear/layout/disklayout.conf' is as you expect.

1) View disk layout (disklayout.conf)  4) Go to Relax-and-Recover shell
2) Edit disk layout (disklayout.conf)  5) Continue recovery
3) View original disk space usage      6) Abort Relax-and-Recover
#? 5
Partition primary on /dev/vda: size reduced to fit on disk.
Please confirm that '/var/lib/rear/layout/diskrestore.sh' is as you expect.

1) View restore script (diskrestore.sh)
2) Edit restore script (diskrestore.sh)
3) View original disk space usage
4) Go to Relax-and-Recover shell
5) Continue recovery
6) Abort Relax-and-Recover
#?
```

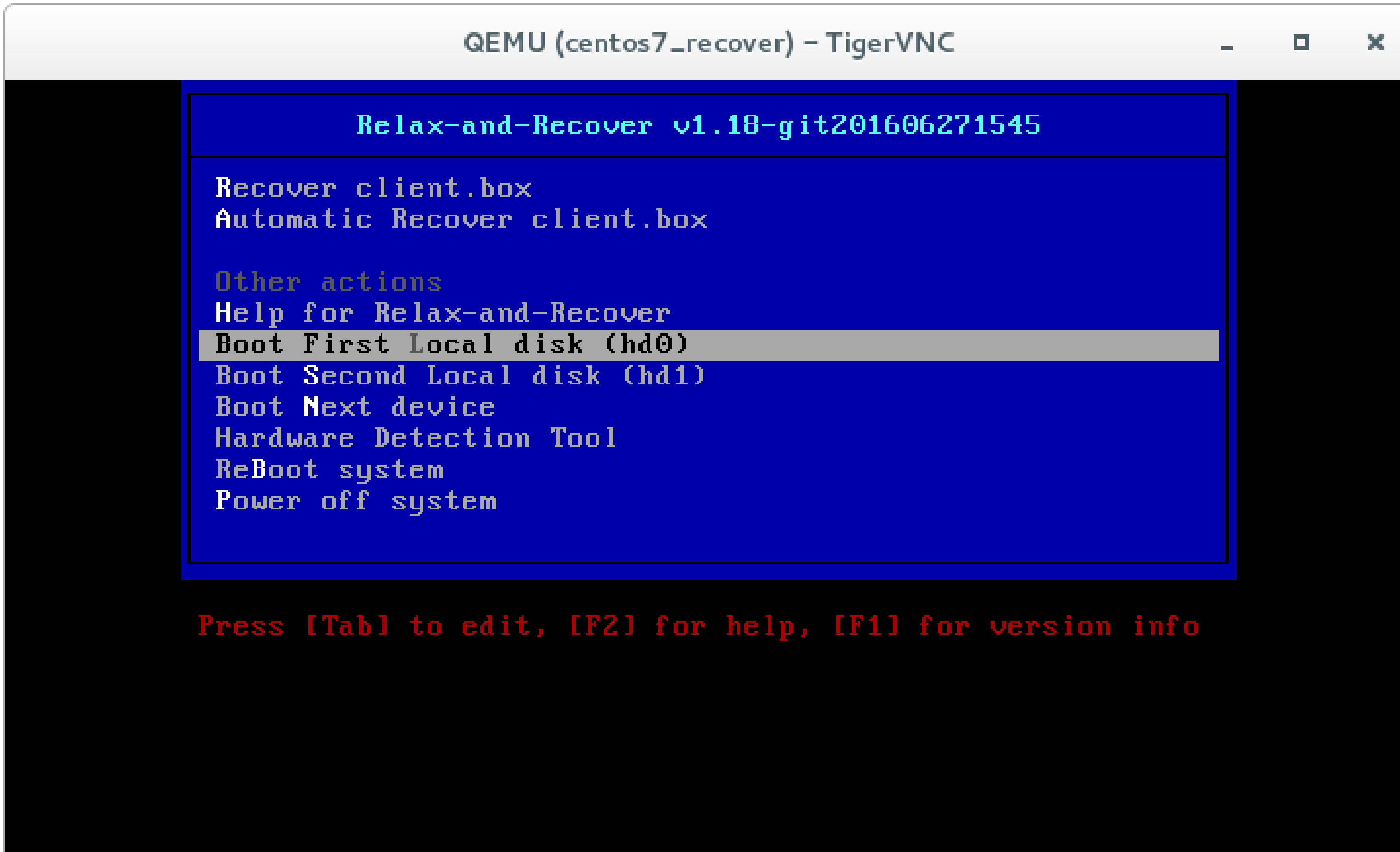
Recover with Bareos (3)

QEMU (centos7_recover) - TigerVNC

```
Catalog:          MyCatalog
Priority:         10
Plugin Options:   *None*
OK to run? (yes/mod/no):
Job queued. JobId=2
You have messages.

waiting for job to start
JobId 2 Job client-restore.2016-09-05_15.17.55_12 is running.
waiting for job to finish
Restore job finished.

Please verify that the backup has been restored correctly to '/mnt/local'
in the provided shell. When finished, type exit in the shell to continue
recovery.

rear> df
Filesystem                      1K-blocks    Used Available Use% Mounted on
devtmpfs                           195988       0    195988   0% /dev
tmpfs                              250420       0    250420   0% /dev/shm
tmpfs                              250420    4248    246172   2% /run
tmpfs                              250420       0    250420   0% /sys/fs/cgroup
/dev/mapper/VolGroup00-LogVol00  38764728 1238724  35928892   4% /mnt/local
/dev/vda2                          487571  195247    262628  43% /mnt/local/boot
rear> _
```

Recover with Bareos (3)

Reboot the VM and select now:



Lab 6

Troubleshoot the configuration files

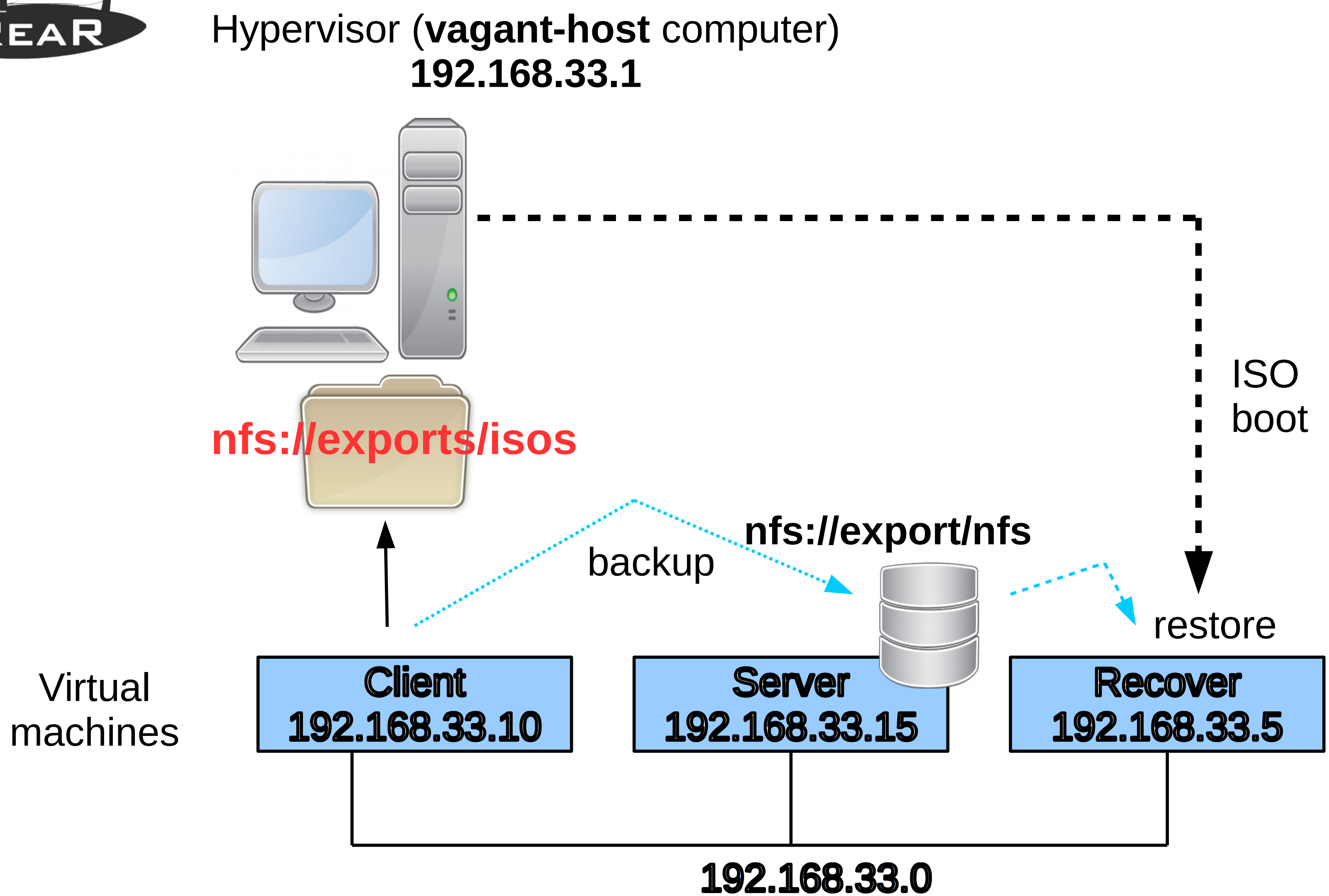
Troubleshoot the configuration files

- Go to directory /etc/rear/workshop
- Type:
```
for f in $(ls *.conf) ; do
    bash -n $f
done
```
- And now type:
```
for f in $(ls *.conf) ; do
    source $f
done
```

Lab 7

Using NFS as backup destination

Lab 7: using NFS



Configure /etc/rear/local.conf

- Replace the content of /etc/rear/local.conf with:

```
OUTPUT=ISO
OUTPUT_URL=nfs://vagrant-host/exports/isos
ISO_DEFAULT=manual
BACKUP=NETFS
BACKUP_URL=nfs://server/export/nfs
USE_STATIC_NETWORKING=y
```

- Or, copy the prepared configuration file:

```
cp /etc/rear/workshop/local-with-nfs.conf /etc/rear/local.conf
```

Using NFS as backup destination

- Run a simulation and understand the difference
  - `rear -s mkrescue`
  - `rear -s mkbackup`
- Run on client: `rear -v mkbackup`
- Halt the client vm: `vagrant halt client`
- Start the recover vm:
  - `vagrant up recover`
  - `vncviewer 127.0.0.1:5993`
  - `Run: rear -v recover`

Lab 8

Using sshfs as backup destination

Configure /etc/rear/local.conf

- Install: `yum -y install sshfs`
- Replace the content of /etc/rear/local.conf with

```
OUTPUT=ISO
OUTPUT_URL=nfs://vagrant-host/exports/isos
ISO_DEFAULT=manual
BACKUP=NETFS
BACKUP_URL=sshfs://root@server/export/archives
USE_STATIC_NETWORKING=y
```

- Or, copy the prepared configuration file:

```
cp /etc/rear/workshop/local-with-sshfs.conf
/etc/rear/local.conf
```

Lab 8 with sshfs

- Make a backup and ISO image (on <u>client</u>):
  - `rear -v mkbackup`
  - NFS mounted via sshfs fuse module
- Halt the client vm
- Start the recover vm:
  - `vagrant up recover`
  - `vncviewer 127.0.0.1:5993`
  - Run: `rear -v recover`

Lab 9

Using cifs as backup destination

Configure /etc/rear/local.conf

- Install: `yum -y install cifs-utils`
- Replace the content of /etc/rear/local.conf with

```
OUTPUT=ISO
OUTPUT_URL=nfs://vagrant-host/exports/isos
ISO_DEFAULT=manual
BACKUP=NETFS
BACKUP_URL=cifs://server/homes
BACKUP_OPTIONS="cred=/etc/rear/.cifs"
USE_STATIC_NETWORKING=y
```

- Or, copy the prepared configuration file:

```
cp /etc/rear/workshop/local-with-cifs.conf
/etc/rear/local.conf
```

Lab 9 with cifs

- `cp /etc/rear/workshop/.cifs to /etc/rear/`
- Make a backup and ISO image (on client):
  - `rear -v mkbackup`
  - Distination mounted via cifs
- Halt the client vm
- Start the recover vm:
  - `vagrant up recover`
  - `vncviewer 127.0.0.1:5993`
  - Run: `rear -v recover`

Need Assistance?

We can foresee in a customized workshop
on consultancy basis
See http://www.it3.be/rear-support

Gratien D'haese
IT3 Consultants